VSL task with established peer work as backbone & minimal adapted dependencies

1. What a suitable backbone must provide

Can an established variable speed limit (VSL) model support the information, control and gradient questions with few traffic-model changes? Consider an ending lane E, its adjacent continuing lane A and farther lane B. Initial states can share the same E and combined A+B traffic at every road position while differing in A/B loading. We ask whether that difference changes merge service, useful common-speed advice and its sensitivity. A suitable backbone must represent the distinction, conserve traffic, respect receiving space and count every initial unit's time until exit. Simplicity concerns the traffic rules we must author and justify, not citation count.

2. What the closest published models offer

Aggregate VSL models support questions about traffic location or history; lane models additionally resolve A/B loading. The inspected starting points offer the following tradeoff.

Table 1. Published starting points and their fit to the question

Published model	What is already supplied	Fit to the research question
Jin and Jin [1]	Spatial and link-average traffic; VSL entry control, capacity drop and upstream holding queues.	Simpler if the question concerns *where* traffic lies along the road. It changes the A/B information question.
Wei, Han and Wang [2]	Link traffic retaining entry/speed history, interacting speed cohorts and capacity drop.	A strong aggregate alternative, but no explicit A/B loading. Native advice acts on entering traffic.
Nagalur Subraveti et al. [3]	Successive lane-cell densities, lane-change incentives and a lane-drop case; Han congestion lineage [4].	Provides the lane backbone; Han's VSL sending restriction [5] and an adapted movement rule complete the task.
Carey et al. [6]	Multilane cell transmission, mandatory movements and receiving-allocation alternatives.	Supports conservative service and forward continuation. As the main framework, it would also need VSL and lane-end behavior specified.

Model construction. The task uses Subraveti's lane framework as the backbone and follows its adoption of Han's congestion approach, completed and adapted for the VSL task. One Carey-led adaptation replaces movement realization; Wright supports its allocation rule. Subraveti's thesis clarifies the backbone. Thus *one principal framework with inherited budgets and one adapted movement component* carries the traffic model. This concentrates independent additions within the compared literature; global minimality is not proved.

Source hierarchy and color key.

Principal framework	**Inherited budget lineage**	**Movement adaptation**
Subraveti article [3]	Han 2016 [4] (adopted in [3])	Carey et al. [6] (principal)
Subraveti thesis [7] (clarification)	Han 2017 [5] (VSL extension)	Wright et al. [8] (supporting)

Comparison only: Jin–Jin [1]; Wei et al. [2]. This task: scenario, composition and task-specific rules.

Source colors identify the work credited; the prose distinguishes retained rules from adaptations. Red: definitions and choices specific to this task, without a novelty claim. Dark gray: standard algebra, conservation and calculus.

3. The formed modeling backbone

3.1. Lane-cell traffic state. The lane-density architecture forms a *lane-resolved cell-transmission-type (CTM) model* with inherited Han budgets. Physical cells retain density and finite storage throughout the road. E can change into A at upstream cell boundaries; traffic in the final E cell must merge at the exit. Unserved mass stays in its current cell, so congestion can extend upstream. The causal chain is:

Lane stocks ⟶ Desires / budgets ⟶ Adapted movements ⟶ New stocks

At step k, density $\rho_{i\ell}^k$ and continuous stock $n_{i\ell}^k = \Delta x \rho_{i\ell}^k$ describe section i, lane $\ell \in \{E, A, B\}$. These labels name current lanes: early-merged E traffic joins A's stock. The task uses $N = 10$ sections, $\Delta x = 0.1\,\text{km}$ and $\Delta t = 1/600\,\text{h}$ (6 s).

3.2. Core modeling assumptions. The task assumes:

- **Cohort and advice.** Initialize a finite cohort inside the road, independently of advice; no later traffic enters. Choose one common $v \in [30, 60]$ km/h at the start and hold it constant.
- **Common lane physics.** Fix the geometry and use one common nominal triangular fundamental diagram (FD), relating flow to density: nominal free speed $u_0 = 60$, backward-wave speed magnitude $w = 20$ km/h, jam density $k_J = 150$ veh/km/lane and maximum capacity-drop fraction $\delta = 0.1$. Initial densities lie in $[0, k_J]$; mass is continuous.
- **Exit and accounting.** Trips end when traffic leaves through A or B; downstream congestion does not block the exit. Count every initial unit's time until it leaves, including traffic still queued at the lane end.
- **Service and updates.** All movements use old traffic and space and advance at most one section per step. Use equal positive priorities and no extra merging-headway factor. The grid satisfies $u_0\Delta t, w\Delta t \le \Delta x$.
- **Sensitivity.** Differentiate the complete objective where the active traffic rules remain unchanged; switching cases need separate treatment.

3.3. Sending and receiving capacity. Movement needs both traffic able to leave and room to accept it. The *sending rate* D is how much a cell can release per hour; the *receiving rate* R is how much the target lane can accept per hour from all competing streams. Both are in vehicle-equivalents per hour. Subraveti adopts Han for capacity drop [3]; we follow that inherited formulation [4] with nominal/VSL caps [5]. The task-specific completion keeps the full density-dependent sending ceiling active, including when the next cell is denser. FD identities are standard. The discharge ceiling $Q(x)$ equals nominal capacity C below critical density ρ_c, then falls linearly to $(1-\delta)C$ at jam density k_J:

$$C = \frac{u_0 w k_J}{u_0 + w}, \quad \rho_c = C/u_0, \quad Q(x) = \min\left\{C, C\left[1 - \delta\frac{x - \rho_c}{k_J - \rho_c}\right]\right\}, \tag{1}$$

$$D_{i\ell} = \min\{v\rho_{i\ell}, Q(\rho_{i-1,\ell})\}, \quad R(x,y) = \min\{Q(x), w(k_J - y), w(k_J - x) + w_d(x - y)\}. \tag{2}$$

Here x, y are the current upstream/downstream densities in the receiving lane. The three receiving bounds express discharge capacity, downstream space and Han's additional congestion-wave constraint. Its fixed recovery slope is $w_d = Q_J/(k_J - Q_J/u_0)$, $Q_J = (1-\delta)C$. The adapted lane-wise budget mapping gives $d_{i\ell} = \Delta t D_{i\ell}$ and $r_{ij} = \Delta t R(\rho_{ij}, \rho_{i+1,j})$, amounts available in one step. All straight and diagonal arrivals to target j share that *one* r_{ij}. At the exit, $r_{Nj} = \Delta t R(\rho_{Nj}, 0)$, $j = A, B$, represents an unblocked downstream road; upstream boundary density is zero. Advice caps sending; receiving changes through the evolving densities, not directly through v.

3.4. Lane-change incentives. This component describes why traffic wants to move over before the lane ends. Subraveti's desire rule [3] compares perceived densities K, combining the current cell and next two cells with weights 2:2:1 [7]. The dimensionless multiplier I expresses the incentive for a particular lane change; $P_{i,\ell j}$ is the desired lateral proportion before shared traffic/service limits are applied:

$$P_{i,\ell j} = \left[\frac{I_{i,\ell j} K_{i\ell} - K_{ij}}{K_{i\ell} + K_{ij}}\right]_+, \qquad I = 1 + I^r + I^{kr} + I^{coop}, \qquad [x]_+ = \max(x, 0). \tag{3}$$

With $I = 1$, only a less crowded destination attracts traffic. A larger multiplier can motivate a change even toward a more crowded lane. The added terms represent route urgency, keep-right preference and cooperation. Within the last $D_r = 0.75$ km, E-to-A route bias grows as $I^r = (1 - g_i/D_r)^3$, where $g_i = (N - i)\Delta x$; it is zero farther upstream. This adds urgency, not a fixed merge fraction or a hard start line: density incentives can cause earlier E-to-A movement. Task-specific incentive conditions determine when and in which direction keep-right and cooperation apply; uncongested A traffic can be encouraged toward B to make room.

3.5. Movement allocation and forced merge. Desire does not guarantee a lane change: requests must share the available traffic and receiving space. Before E ends, a failed attempt may still continue forward in E; at its end, traffic must merge or wait. The adapted movement rule [6] replaces Subraveti's within-section lateral moves and adjusted forward-flow coupling. At each section, ℓ is the origin lane, j the destination lane. The desired lateral proportion $P_{\ell j}$ becomes an assigned fraction $b_{\ell j}$ of the origin's sending amount d_ℓ; $m_{\ell j}$ is the resulting request. Proportional normalization preserves desires whose sum is at most one, otherwise scales them to fit one shared sending amount. The remainder goes straight. Two 70% lateral requests therefore become 50% each, rather than spending 140% of the same traffic:

$$b_{\ell j} = \frac{P_{\ell j}}{\max(1, \sum_{h \neq \ell} P_{\ell h})} \ (j \neq \ell), \quad b_{\ell\ell} = 1 - \sum_{j \neq \ell} b_{\ell j}, \quad m_{\ell j} = d_\ell b_{\ell j}. \tag{4}$$

We adapt Carey's proportional receiving rule [6] to the sender's budget. The straight remainder is standard accounting; request formation follows [3, Eq. (14)]. Carey's priority sharing and forward continuation [6] underpin receiving allocation. We extend its two-stream sharing to all target requests; Wright's priority formulation [8] provides supporting justification, without importing its general junction model.

For each target, choose the common service level λ_j so that the served amounts $z_{\ell j}$ fill its receiving amount r_j, or satisfy all requests if they are smaller:

$$z_{\ell j} = \min(m_{\ell j}, \lambda_j), \qquad \sum_{\ell} z_{\ell j} = \min\left(r_j, \sum_{\ell} m_{\ell j}\right). \tag{5}$$

Small requests are fully served and release unused entitlement to the others. Forward continuation lets traffic not served in that allocation still advance in its own lane if space remains. This additional flow y_ℓ is limited by unused sending and unused same-lane receiving. If receiving space also prevents forward movement, the traffic waits in its current cell. Thus $f_{\ell j} = z_{\ell j} + \mathbf{1}_{\{j=\ell\}} y_\ell$ is realized movement. An unsuccessful E-to-A attempt can therefore continue in E if space remains. Final-E rule: $b_{EA} = 1$, $y_E = 0$; E must use A's outlet or wait. It competes with A and any B-to-A request for the same outlet budget. Equal priorities are not a constant realized merge proportion; demand and receiving state determine the flow. At the choke, E uses part of A's exit capacity, so A's own traffic may have to wait. The resulting accumulation in A can reduce space for later arrivals and propagate congestion upstream. Merge effects enter through shared service and density evolution; speed differences and acceleration recovery are not modeled separately. Normalization and three-lane sequencing are behavioral adaptations in addition to the lane-end geometry change.

Adapted diagonal transport advances one section when changing lanes. Apply old-state flows simultaneously using standard conservation:

$$n_{i\ell}^{k+1} = n_{i\ell}^{k} - \sum_{j} f_{i,\ell j}^{k} + \sum_{h} f_{i-1,h\ell}^{k}. \tag{6}$$

Set $f_0 = 0$; f_N counts traffic leaving the modeled road. Matched additions/subtractions conserve mass, while donor/receiver limits protect stock and storage.

3.6. Total system time and sensitivity. Minimize total system time J: all initial traffic's time until exit. Once remaining mass is below $\Delta x \rho_c$, all requests receive service and a unit in section i has remaining mean time $(N - i + 1)\Delta x / v$. Choose a fixed index K_* inside that regime. The weight $h_{i\ell} = N - i + 1$ counts the remaining cell advances from section i to the exit; $\mathbf{1}^\top n^k$ is the mass still on the road. The complete objective is

$$J(v) = \Delta t \sum_{k<K_*} \mathbf{1}^\top n^k + \frac{\Delta x}{v} h^\top n^{K_*}. \tag{7}$$

Population-time accounting is standard; the exact residual is derived for this task. The first term adds the time already spent by all unfinished traffic before the cutoff. The second adds all travel time still to come: each remaining stock is weighted by its distance to the exit and divided by advice speed. Traffic remaining at the cutoff is therefore counted rather than discarded. Write the complete update as $n^{k+1} = F(n^k, v)$. Between branch switches, $s^0 = 0$, $s^{k+1} = F_n^{[k]} s^k + F_v^{[k]}$ propagates $s^k = \partial n^k / \partial v$ through budgets, desires, allocation and stock. Here F_n, F_v are the update's first derivatives with respect to state and advice. Then

$$J'(v) = \Delta t \sum_{k<K_*} \mathbf{1}^\top s^k + \frac{\Delta x}{v} h^\top s^{K_*} - \frac{\Delta x}{v^2} h^\top n^{K_*}. \tag{8}$$

This model's full derivative uses the standard chain rule. Terms measure earlier traffic changes, residual-stock changes and direct residual travel time. Hold a valid K_* fixed locally; at traffic-regime or active-constraint changes a two-sided derivative can fail.

3.7. Validation design and rival models. The validation design asks: does hiding A/B loading worsen advice, can one local reading suffice, and do gradients help choose advice? The planned comparisons keep the physical model and complete objective fixed:

1. Isolate information value. Compare an observer seeing E and A+B at each section with one seeing each lane. For paired states with identical reduced observations but different A/B loading, the reduced observer must choose one shared advice; the full observer may choose separately. Allow model-based inference from the available observation history; decide before later discharge reveals the split.

2. Test a compact information rival. Add only final A-cell density to E and A+B. Its value, 60 or 90 veh/km/lane, identifies which of the two initial states is present. Compare with full information to test whether one reading suffices; sufficiency beyond that pair is unproved.

3. Check gradients, then their usefulness. First check Section 3.6's complete-objective derivative against finite differences away from rule switches. Then, within each information view and with matched initialization and computational budget, compare gradient-guided advice with direct speed search and maximum advice. Search is a strong scalar-action reference, not a certified global optimum; maximum advice tests benefit over simply using the highest speed. This separates missing information from poor optimization. Numerical validation remains to be carried out.

References

[1] H.-Y. Jin and W.-L. Jin, "Control of a lane-drop bottleneck through variable speed limits," arXiv:1310.2658, 2013. [Online]. Available: `https://arxiv.org/abs/1310.2658`.

[2] L. Wei, Y. Han, and M. Wang, "A first-order link-based flow model with variable speed limits and capacity drops for freeway networks," *Comput.-Aided Civ. Infrastruct. Eng.*, vol. 40, no. 21, pp. 3200–3217, 2025, doi: 10.1111/mice.13492.

[3] H. H. S. Nagalur Subraveti, V. L. Knoop, and B. van Arem, "First order multi-lane traffic flow model—an incentive based macroscopic model to represent lane change dynamics," *Transportmetrica B, Transp. Dyn.*, vol. 7, no. 1, pp. 1758–1779, 2019, doi: 10.1080/21680566.2019.1700846.

[4] Y. Han, Y. Yuan, A. Hegyi, and S. P. Hoogendoorn, "New extended discrete first-order model to reproduce propagation of jam waves," *Transp. Res. Rec.*, vol. 2560, pp. 108–118, 2016, doi: 10.3141/2560-12.

[5] Y. Han, A. Hegyi, Y. Yuan, S. Hoogendoorn, M. Papageorgiou, and C. Roncoli, "Resolving freeway jam waves by discrete first-order model-based predictive control of variable speed limits," *Transp. Res. C, Emerg. Technol.*, vol. 77, pp. 405–420, 2017, doi: 10.1016/j.trc.2017.02.009.

[6] M. Carey, C. Balijepalli, and D. Watling, "Extending the cell transmission model to multiple lanes and lane-changing," *Netw. Spat. Econ.*, vol. 15, pp. 507–535, 2015, doi: 10.1007/s11067-013-9193-7.

[7] H. H. S. Nagalur Subraveti, *Lane-specific Traffic Flow Control*, Ph.D. dissertation, Delft Univ. Technol., Delft, The Netherlands, 2021, ch. 4, doi: 10.4233/uuid:e405a549-b1fe-4fa1-8090-4429612359d9.

[8] M. A. Wright, G. Gomes, R. Horowitz, and A. A. Kurzhanskiy, "On node models for high-dimensional road networks," *Transp. Res. B, Methodol.*, vol. 105, pp. 212–234, 2017, doi: 10.1016/j.trb.2017.09.001.